ZOMATHON 2026 · PRODUCT NOTE

BITE BOOST Intelligent Add-On Recommendation Engine

Production-grade ML system replacing static popularity lists with personalized, context-aware add-on suggestions at checkout — driving measurable incremental revenue per order.

- **+35%** ACCEPTANCE
- **+₹26** AOV UPLIFT
- **<1.1ms** P95 LATENCY

THE PROBLEM

STATIC RANKINGS
Same suggestions for every user regardless of cart

NO PERSONALIZATION
Ignores time, cuisine preference, and history

COLD-START GAP
New users (<5 orders) get poor recommendations

NO CART AWARENESS
Doesn't adapt as items are added to cart

OUR SOLUTION — 5 STRATEGIC LAYERS

ML Ranking
LightGBM on 845K pairs, real-time scoring

Segment Intel
High-value vs casual cohort optimization

Cold-Start Fix
Hybrid fallback: 42%→61% new users

Sequential Cart
Multi-round re-rank after each accept

Real-Time
p95 <1.1ms, offline→online serve

MEASURED IMPACT — BASELINE VS MODEL

+35% PRIMARY
ACCEPTANCE RATE LIFT
52.9% → 71.7%

₹830
AVG ORDER VALUE
↑ from ₹804 · +₹26/order

+46%
COLD-START LIFT
42% → 61% for new users

73.5%
HIGH-VALUE ACCEPTANCE

1.08ms
P95 LATENCY

3 rounds
SEQUENTIAL SIMULATION

BASELINE 52.9% —(+35%)— 71.7% MODEL

VALIDATION & METHODOLOGY

- Temporal split to prevent leakage (train earlier dates, test later dates)
- Evaluate ranking quality at K (Precision@10 / Recall@10 / NDCG@10)
- Segment checks (High-value vs Casual)
- Cold-start check (new users vs warm users)
- Error review: inspect underperforming segments and adjust features

PROJECTION

₹26 uplift per order

₹2.6 Cr/day at 1M orders/day

₹0.80 Cr/day at 30% adoption (conservative)

A/B TEST PLAN

A/B design:
Control: Popularity baseline (current-style suggestions)
Treatment: ML ranker + guardrails

Primary metrics:
Add-on Acceptance Rate, Avg Order Value (AOV), Cart-to-Order (C2O)

Guardrails:

- Latency SLA (end-to-end <200ms; model scoring p95 = 1.08ms)
- Cart abandonment rate must not worsen
- No duplicate suggestions / diet preference respected
- Price-shock cap to protect UX

SYSTEM ARCHITECTURE

Cart → Features → **LightGBM** (845K samples) → Ranked K → UI

feedback loop

Offline train → Online serve at checkout → Accept/reject feeds back into **periodic retraining**

WHY THIS WINS

- Business-first thinking
- Real-world deployable
- Cold-start handling
- Segment targeting
- Latency benchmarked
- Modular codebase
- Sequential cart logic
- Production-ready

zomato
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github.com/AshwinSheoran02/zomato-recommendation-system

Submission links (public): GitHub repo · dataset.zip · reproducible scripts/notebooks

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+₹26 per order

845K samples · 1.08 ms p95 · Production-ready simulation

Resources (public): GitHub repo, dataset.zip, reproducible scripts.